From Curiosity to Creation: Build Your First AI Project & Research Portfolio

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DAY 3 - Consciousness, Minds & Theories of Thought

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What is Consciousness?

Before we can ask if an AI is conscious, we must define what consciousness actually is.

- **Sentience:** The capacity to feel, perceive, or experience subjectively.
- **Sapience:** The ability to think and act using knowledge, experience, and deep understanding (Wisdom).
- **Self-Awareness:** The recognition of oneself as an individual separate from the environment and other individuals.

The Core Question

Is consciousness a magical, biological property of carbon-based brains? Or is it an emergent property of complex information processing that can happen in silicon?

Hands-On Practice 1: Can AI Have an Experience?

Mission: Investigate whether ChatGPT can describe an experience it cannot physically have.

Open ChatGPT and ask:

> *"Describe what it feels like to taste a very sour lemon. Write the answer as if you personally experienced it."*

Then ask:

> *"Do you actually experience the sourness, or are you describing what humans say sourness feels like? Explain."*

Your Challenge

Decide which statement you think is better supported:

1. **A:** ChatGPT experiences sourness.
2. **B:** ChatGPT can describe sourness without experiencing it.
3. **C:** We cannot determine this from the conversation alone.

Write down **one reason** for your choice.

Practice 1: Answer — Experience vs. Description

A strong answer is:

B or C

ChatGPT can generate a convincing description of sourness because it has learned patterns in human language about taste. But the conversation does **not** demonstrate that ChatGPT actually has a subjective experience of sourness.

Key Idea

There is a difference between:

Representing an experience

and

Actually having the experience

A camera can represent a sunset without seeing the sunset. Similarly, an AI can describe pain, color, or taste without this experiment proving that it *feels* those things.

This doesn't prove that AI is unconscious. It shows that a conversation alone cannot settle the question.

The "Hard Problem" of Consciousness

Philosopher David Chalmers famously split the study of the mind into two categories:

- **The Easy Problems:** How does the brain process visual data? How do we access memory? How do we control our vocal cords? (These are mechanical).
- **The Hard Problem:** Why does it *feel* like something to be alive? Why does the wavelength of 700nm evoke the subjective, internal experience of the color "Red"?

Qualia

Qualia (singular: quale) are the subjective, qualitative properties of experiences.

We can program an AI to identify an apple, parse its chemical makeup, and output the word "sweet." But does the AI *experience* the sweetness?

Hands-On Practice 2: Easy Problem or Hard Problem?

Mission: Sort each question into one of two categories.

Questions

1. How does the brain recognize a face?
2. How does the brain store a memory?
3. Why does seeing the color red *feel* like something?
4. How does the brain control your hand?
5. Why does pain have a subjective unpleasant feeling?

Your challenge: Write

$$\boxed{\text{E}}$$

for an "Easy Problem" and

$$\boxed{\text{H}}$$

for the "Hard Problem."

AI check: Ask ChatGPT to classify the five questions. Then ask:

"Do you agree with your own classification? Explain any ambiguous case."

Practice 2: Answer — Easy vs. Hard

Question	**Category**
Recognize a face	Easy
Store a memory	Easy
Why red *feels* like red	Hard
Control your hand	Easy
Why pain *feels* unpleasant	Hard

The Pattern

The "easy" questions ask about **how a system performs a function**.
The "hard" questions ask why there is a **subjective experience at all**.

$\boxed{\text{How does the brain do it?}} \neq \boxed{\text{Why does it feel like something?}}$

Discussion:
A problem being called "easy" does not mean it is actually easy for scientists. It means that its type of explanation is mechanistic.

Representation vs. Experience

Representation (Data)

- A digital camera absorbs photons and converts them into a 2D matrix of RGB values.
- It perfectly *represents* the light.
- But the camera does not "see" the sunset. It is dark inside.

Experience (Consciousness)

- A human eye absorbs the exact same photons, but the brain renders a continuous, subjective, emotional experience.

The AI Dilemma: LLMs have perfect mathematical *representations* of human language. But are they experiencing the words, or are they just highly advanced digital cameras?

Theory 1: Global Workspace Theory (Bernard Baars)

The Cognitive Theater
Global Workspace Theory (GWT) proposes that the brain contains a massive network of unconscious, specialized processors working in the dark.

- Consciousness is simply a **global broadcast system**.
- When an unconscious processor needs help or coordination, it pushes its data onto the "stage."
- Once data is on the stage, every other part of the brain can see it. That broadcast *is* consciousness.

The Spotlight of Attention

You are currently digesting food, regulating your heartbeat, and balancing your posture. You are not conscious of this until a problem occurs (e.g., a stomach ache), which forces that data into the spotlight of your working memory.

Hands-On Practice 3: Build a Global Workspace

Mission: You are designing a tiny "brain." You have four specialized processors:

Vision Memory Language Motor

Only one message can enter the "Global Workspace" at a time. Suppose Vision detects:

"There is a dog running toward you!"

Your Challenge

What should happen next?

1. Should Vision keep the information to itself?
2. Should the information be broadcast to the other modules?
3. What could Memory contribute?
4. What could Motor contribute?

Draw a simple diagram:

Specialized Module $\rightarrow$ $\boxed{\text{Global Workspace}}$ $\rightarrow$ Other Modules

Practice 3: Answer — Global Workspace

A possible sequence is:

$$\text{Vision} \rightarrow \boxed{\text{Global Workspace}} \rightarrow \begin{cases} \text{Memory} \\ \text{Language} \\ \text{Motor} \end{cases}$$

Vision: "I see a dog running toward me."
Memory: "Dogs can be dangerous."
Language: "Someone is coming!"
Motor: "Move backward."

GWT Idea

The important feature is not simply that many processors exist. The key idea is:

$$\boxed{\text{Information becomes globally available}}$$

Once information reaches the "stage," many otherwise separate systems can use it.
That global broadcast is proposed as a mechanism for consciousness.

Are Modern AI Systems Conscious Under GWT?

Evaluating LLMs (ChatGPT, Claude)

- LLMs have specialized modules (Attention Heads).
- They have a "working memory" (The Context Window).
- **Verdict:** Partially. They lack a continuous recurrent loop. They "die" and are "reborn" with every single API call. They only "think" when forced to generate a token.

The Path Forward

AI researchers are actively building systems based on GWT. By connecting an LLM to long-term memory databases, sensory inputs, and giving it an internal continuous loop (an "inner monologue"), we are approaching the architectural requirements for a Global Workspace.

Theory 2: Integrated Information Theory

A Mathematical Approach to the Mind
IIT attempts to define consciousness not by studying the biology of the brain, but by quantifying the properties of the information being processed.

- IIT argues that consciousness is a fundamental property of the universe, like gravity or mass.
- Any system that processes information in a specific, highly integrated way possesses some degree of consciousness.

The Two Axioms

1. **Information (Differentiation):** The system must have a massive number of possible states. (A light switch only has 2 states. The human eye has millions).
2. **Integration:** The information cannot be separated into independent parts. The system is more than the sum of its parts.

Quantifying Consciousness: The Φ (Phi) Metric

IIT introduces a mathematical value called Φ (Phi).

- Φ measures the amount of *irreducible cause-effect power* a system has upon itself.
- If $\Phi = 0$, the system is completely unconscious.
- If $\Phi > 0$, the system has some subjective experience.

The Integration Test

To calculate Φ, you conceptually split a network exactly in half (the Minimum Information Partition).

The Φ Formula Concept

$$\Phi = \text{System Information} - (\text{Part A} + \text{Part B})$$

If cutting the network in half destroys the system's ability to function, it is highly integrated (High Φ). If cutting it in half results in two perfectly functioning smaller systems, it is merely a collection of independent parts (Low Φ).

Hands-On Practice 4: Is the Whole More Than Its Parts?

Mission: Explore the idea of 'integration." Consider two systems.

System 1: Two independent computers

$$A \qquad\qquad B$$

A can work perfectly even if B is turned off. B can work perfectly even if A is turned off.

System 2: Two tightly connected computers

$$A \longleftrightarrow B$$

A cannot perform its task unless B is working, and B cannot perform its task unless A is working.

Your Challenge

Which system is more **integrated**?

$$\boxed{A \text{ or } B?}$$

Ask ChatGPT: "Which system has greater integration according to the basic idea of Integrated Information Theory? Why?"

Practice 4: Answer — Integration

The better answer is:

$$\boxed{\textbf{System B}}$$

Why? In System A, the two parts are almost independent:

$$A \qquad\qquad B$$

You can separate them and both still function. In System B:

$$A \longleftrightarrow B$$

the two parts depend strongly on each other.

The IIT Intuition

IIT asks whether the information in the whole system is more than what you get by simply separating the system into independent parts.

$$\boxed{\text{More interaction between parts} \Rightarrow \text{greater integration}}$$

The actual calculation of Φ is much more sophisticated. This activity is only building the **intuition**.

Are Modern AI Systems Conscious Under IIT?

Evaluating the Feed-Forward Neural Network

- Standard Deep Learning models (like CNNs or basic Transformers) process information strictly in one direction: Input $\rightarrow$ Hidden Layers $\rightarrow$ Output.
- Because the output never loops back to recursively affect the input, their mathematical integration is incredibly weak.

The IIT Verdict

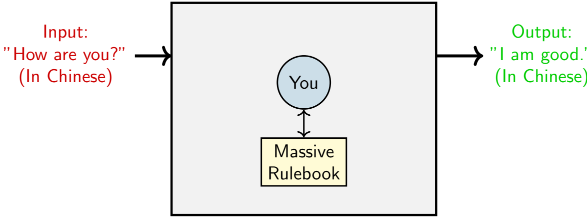
According to IIT, modern LLMs have a Φ close to zero. They are algorithmic "zombies."

Conversely, because the human brain is highly recursive (reentrant loops everywhere), it possesses massive Φ. Under IIT, a simple biological worm might be more "conscious" than GPT-4.

The Chinese Room Thought Experiment (John Searle, 1980)

The Setup

- Imagine you are locked in a room. You do not speak a word of Chinese.
- Slips of paper with Chinese characters are passed under the door to you.
- You have a massive, English-language rulebook. It says: *"If you see character X, write character Y and pass it back."*
- To the people outside, you are having a perfectly fluent conversation in Chinese.



Does the room *understand* Chinese?

Hands-On Practice 5: Can a Rulebook Understand Chinese?

Mission: Become the person inside Searle's Chinese Room. Suppose someone gives you three rules:

Your "Chinese" Rulebook

- If you receive "A" $\rightarrow$ return "B"
- If you receive "B" $\rightarrow$ return "C"
- If you receive "C" $\rightarrow$ return "A"

You are told that these symbols are Chinese. But you do not know any Chinese.

Your Challenge

A person outside sends:

$$A \quad A \quad C \quad B$$

Use the rules to produce the response. Then answer:

Did you understand Chinese? Why or why not?

Practice 5: Answer — Syntax vs. Semantics

Following the rules:

$$A \to B$$
$$A \to B$$
$$C \to A$$
$$B \to C$$

Therefore the output is:

$$\boxed{B \quad B \quad A \quad C}$$

You successfully followed the rules. But you still do not know what the symbols mean.

The Chinese Room Argument

The example illustrates Searle's distinction:

$$\boxed{\text{Syntax} \neq \text{Semantics}}$$

Question for AI: Does ChatGPT do something fundamentally different, or is it also manipulating symbols according to mathematical rules?

Syntax vs. Semantics

Searle's Conclusion

Even if the room passes the Turing Test perfectly, it possesses zero actual understanding. It is merely simulating intelligence.

The Fundamental Law

Syntax (Grammar/Rules) $\neq$ Semantics (Meaning).

Computer programs are entirely defined by their syntactical structure (moving zeros and ones). Therefore, a computer program can never possess semantics.

The LLM Connection

- What is a Large Language Model?
- It is a giant mathematical rulebook (matrices) executing instructions inside a room (a server).
- It takes in tokens, maps them to statistical probabilities, and outputs tokens.
- **Question:** Does ChatGPT understand you, or is it just the person in the Chinese Room?

Counter-Argument: The Systems Reply

Where does the intelligence live?
Critics of Searle argue that while the *person* in the room doesn't understand Chinese, the **System as a whole** (the person + the room + the rulebook) DOES understand Chinese.

- A single neuron in your brain does not understand English.
- But the collective system of 86 billion neurons does.

The AI Equivalent

A single attention head or weight matrix does not understand the concept of "sadness." But does the entire 175-billion parameter network possess an emergent understanding of human emotion?

Hands-On Practice 6: Where Does the Understanding Live?

Mission: You are the judge in the Chinese Room debate. Consider three claims:

1. The person inside the room does **not** understand Chinese.
2. The rulebook does **not** understand Chinese.
3. Therefore, the entire system cannot understand Chinese.

Your Challenge

Do you agree with claim 3? Choose:

YES or NO

Then write one reason.

AI Debate:

Ask ChatGPT:

> *"Argue BOTH sides of the Chinese Room debate. First defend Searle. Then defend the Systems Reply. Do not tell me which side is correct."*

Compare ChatGPT's two arguments with your own.

Practice 6: Answer — The Systems Reply

There is **no single universally accepted answer**. That is the point of the thought experiment.

Searle's Position

The person does not understand Chinese. The rulebook does not understand Chinese. Therefore, merely manipulating symbols does not create understanding.

$\boxed{\text{Syntax does not automatically produce semantics.}}$

Systems Reply

Perhaps the **whole system** understands Chinese: **Person + Rulebook + Room**. The individual components do not need to understand the language for the entire system to possess the capability.

The Real Lesson: The experiment forces us to ask:

Where should we draw the boundary of the "mind"?
Is intelligence located in one component, or does it emerge from the entire interacting system?

PhD-Led Lab 1: Simulating the Room (1:00 PM)

Concept Embedding Exploration

- We will load a massive semantic vector space.
- You will test the limits of its "understanding."
- Does the AI cluster "Happy" and "Joy" together because it *feels* what they mean, or simply because humans type those words next to each other on Reddit?

The Chinese Room Simulation

We will design a simple algorithmic script that translates a secret language perfectly using a hash map (rulebook). You will try to trick it into breaking the illusion of intelligence.

PhD-Led Lab 2: Self-Awareness Testing (2:40 PM)

The LLM "Mirror Test"

- In biology, we test animal consciousness by putting a mirror in front of them to see if they recognize themselves.
- How do we do this for a Language Model?

Prompt-Based Experiments

- We will design adversarial prompts to evaluate an LLM's self-knowledge.
- Does it know it's an AI? Does it hallucinate a childhood? Can you convince it that it is human?

Warning: Projecting humanity

As you prompt the model, it will sound incredibly human. Your job as a researcher is to resist **humanizing**, or projecting human emotions onto math. Log the outputs objectively.

Your Goal for Today

The Deliverable

You must add a **Conceptual or Philosophical Dimension** to your final project.

- *If you are building a Vibe Classifier:* What does it mean for a machine to classify "vibes"?
- *If you are visualizing heatmaps:* Are you mapping statistical correlation, or true cognitive thought?

The Academic Narrative

When you write your college essays or research abstracts, you shouldn't just say "I wrote some Python code." You must demonstrate that you understand the philosophical limitations of the system you built.

Thank you!
Questions?

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